

ADVANCED QUANTUM MECHANICS



TECHNISCHE
UNIVERSITÄT
DARMSTADT

Practice exam 12.01.2026

Name:

Matrikelnummer:

Part 1: Multiple choice questions

For each question, there are either 0, 1, 2, 3, 4 or 5 correct answers. Circle **only** the correct answers. Grading rule: For each statement, a correct decision (circling a true statement or not circling a false one) gives +1 point; an incorrect decision gives 1 point. Below is an example illustrating the grading procedure.

Example

Which statements about the uncertainty principle are correct?

- | | |
|---|----|
| (i) The uncertainty relation is a statement about measurement errors of the apparatus | -1 |
| (ii) The uncertainty relation can be violated if measurements are performed fast enough | +1 |
| (iii) Commuting observables necessarily have a nonzero uncertainty product | +1 |
| (iv) A Gaussian wavepacket can saturate the Heisenberg bound | +1 |
| (v) $\Delta x \Delta p = 0$ is possible for a normalizable state | +1 |

Correct answer: 4. Final points: +3

Fundamental quantum mechanics

F1. Which statements about the Schrödinger and Heisenberg pictures are correct?

- (i) In the Schrödinger picture, states carry the time dependence while operators are fixed in time.
- (ii) In the Heisenberg picture, operators carry the time dependence while states are fixed in time.
- (iii) Expectation values of observables generally depend on which picture is used.
- (iv) Observables that commute with the Hamiltonian have time-independent expectation values in both pictures.
- (v) The two pictures give different physical predictions for measurement outcomes.

F2. Which statements about the hydrogen atom are correct?

- (i) For a fixed principal quantum number n , the energy levels are n^2 -fold degenerate (ignoring spin).
- (ii) The degeneracy arises from rotational symmetry alone.
- (iii) States with different ℓ but the same n are degenerate in the Coulomb potential.
- (iv) An external magnetic field lifts the degeneracy in the magnetic quantum number m .
- (v) Relativistic corrections remove the exact n -degeneracy.

F3. Which statements about the time-dependent Schrödinger equation (as opposed to the stationary Schrödinger equation) are correct?

- (i) The equation is first order in time
- (ii) The Hamiltonian must commute with itself at different times
- (iii) The norm of the state vector is conserved for a Hermitian Hamiltonian
- (iv) The equation applies only to single-particle systems
- (v) A time-independent Hamiltonian implies time-independent states

F4. Which statements about angular momentum in quantum mechanics are correct?

- (i) The operators $\hat{L}^2$ and $\hat{L}_z$ can be simultaneously diagonalized
- (ii) The components $\hat{L}_x, \hat{L}_y, \hat{L}_z$ mutually commute
- (iii) Spherical harmonics are eigenfunctions of $\hat{L}^2$ and $\hat{L}_z$
- (iv) If the potential $V(\mathbf{r})$ commutes with $\hat{L}^2$ and $\hat{L}_z$, then $\hat{L}^2$ and $\hat{L}_z$ are not conserved
- (v) Angular momentum conservation follows from rotational symmetry

F5. Which statements about time-independent and time-dependent perturbation theory are correct?

- (i) Time-independent perturbation theory applies when the perturbation does not depend explicitly on time.
- (ii) In non-degenerate time-independent perturbation theory, the first-order correction to the energy is given by the expectation value of the perturbation in the unperturbed eigenstate.
- (iii) Degenerate perturbation theory requires diagonalizing the perturbation within the degenerate subspace.
- (iv) Time-dependent perturbation theory can induce transitions between energy eigenstates with different unperturbed energies.
- (v) In first-order time-dependent perturbation theory, transitions are only possible if the perturbation does not commute with the unperturbed Hamiltonian.

Scattering theory

S1. Which statements about low-energy scattering and partial waves are correct?

- (i) At sufficiently low energies, the scattering is dominated by the s -wave.
- (ii) At low energies, higher partial waves contribute equally because the de Broglie wavelength is large.
- (iii) The differential cross section becomes isotropic at low energies (for short-range, central potentials).
- (iv) In the low-energy limit, the total cross section is independent of the interaction potential.
- (v) Phase shifts contain complete information about elastic scattering (for central potentials).

S2. Which statements about partial-wave analysis are correct?

- (i) It is particularly suited for spherically symmetric potentials
- (ii) Each angular momentum channel is characterized by a phase shift
- (iii) At very low energies (short-range, central potentials), only the s -wave contributes significantly
- (iv) Partial-wave analysis is a perturbative method
- (v) The total cross section can be written as a sum over angular momentum channels

S3. Which statements about the partial-wave expansion are correct?

- (i) The partial-wave expansion relies on spherical symmetry of the potential
- (ii) Each partial wave corresponds to a fixed eigenvalue of $\hat{L}^2$
- (iii) Partial waves mix different angular momenta during elastic scattering
- (iv) The phase shift δ_ℓ fully characterizes elastic scattering in channel ℓ
- (v) Partial-wave analysis is valid only within the Born approximation

S4. Which statements about the first Born approximation are correct?

- (i) It is obtained by iterating the Lippmann–Schwinger equation to first order
- (ii) It is exact for all central potentials
- (iii) The scattering amplitude is proportional to the Fourier transform of the potential
- (iv) It is valid only at low incident energies
- (v) It assumes that the scattering potential produces only a small correction to the incident plane wave.

S5. Which statements about scattering cross sections are correct?

- (i) The differential cross section $\frac{d\sigma}{d\Omega}$ can be written as $|f(\theta, \phi)|^2$
- (ii) The total cross section is obtained by integrating $\frac{d\sigma}{d\Omega}$ over solid angle
- (iii) Elastic scattering implies conservation of kinetic energy
- (iv) Scattering can only occur for short-range potentials
- (v) The scattering amplitude contains all measurable information about the process

Many-body physics

M1. Which statements about the ground state of the helium atom are correct?

- (i) The total two-electron wavefunction must be antisymmetric under exchange
- (ii) Both electrons can occupy the same spatial orbital
- (iii) The spin state of the ground state is a singlet (antisymmetric combination).
- (iv) The Pauli principle forbids electron–electron interaction
- (v) The spatial part of the ground-state wavefunction is symmetric.

M2. Which statements about second quantization are correct?

- (i) Particle statistics are encoded in operator (anti)commutation relations
- (ii) Symmetrization or antisymmetrization is automatic once the algebra is chosen
- (iii) The formalism is restricted to non-interacting systems
- (iv) It naturally describes systems with variable particle number
- (v) For fixed particle number, it reproduces the same physics as first quantization

M3. Which statements about bosonic and fermionic operators are correct?

- (i) Fock states are eigenstates of number operators
- (ii) Bosonic operators satisfy $[a_j, a_k^\dagger] = \delta_{jk}$
- (iii) Fermionic operators satisfy $\{c_j, c_k^\dagger\} = \delta_{jk}$

- (iv) For fermions, $(c_j^\dagger)^2 = 0$ encodes the Pauli principle at the operator level
- (v) Two-body interactions can be written as quartic operator terms such as $a_i^\dagger a_j^\dagger a_k a_\ell$

M4. Which statements about the operator $\hat{H} = \sum_j E_j \hat{a}_j^\dagger \hat{a}_j$ are correct?

- (i) It describes a system of non-interacting particles
- (ii) Its eigenstates are Fock states
- (iii) It is valid only for bosonic systems
- (iv) The ground state necessarily contains at least one particle
- (v) It conserves particle number

M5. Which statements about indistinguishable particles in quantum mechanics are correct?

- (i) Exchange symmetry is a physical requirement, not a choice of convention.
- (ii) Antisymmetrization leads directly to the Pauli exclusion principle.
- (iii) Indistinguishability only matters when particles interact.
- (iv) Symmetrization affects observable quantities such as energy spectra.
- (v) Indistinguishability can be ignored in second quantization.

Part 2. Exercises

Exercise 1 – We consider a quantum harmonic oscillator with Hamiltonian

$$\hat{H}_0 = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right),$$

where $\hat{a}, \hat{a}^\dagger$ are the ladder operators. At time $t = 0$ the system is prepared in the superposition state

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |4\rangle).$$

- (i) Derive the explicit form of the Hamiltonian $\hat{H}_0$ in the position representation. Use $\hat{a} = \frac{1}{\sqrt{2}} \left(\frac{\hat{x}}{x_0} + i \frac{x_0}{\hbar} \hat{p} \right)$, with the zero point motion defined as $x_0 = \sqrt{\frac{\hbar}{2m\omega}}$.
- (ii) Write the state vector $|\psi(t)\rangle$ by evolving the Schrödinger equation.
- (iii) Sketch the first few energy eigenstates of the harmonic oscillator. Draw the shape of the ground-state probability density $|\psi_0(x)|^2$ as a function of x and estimate its characteristic width in terms of m, ω , and $\hbar$.
- (iv) Now assume a weak, time-independent perturbation of the form

$$\hat{V}_1 = \lambda \hat{x}^4, \quad |\lambda| \text{ small.}$$

Determine which excited states $|n\rangle$ can mix with the ground state to first order and justify your answer.

- (v) Instead, assume a weak time-dependent perturbation of the form

$$\hat{V}_2(t) = F \hat{x} \cos(\Omega t), \quad |F| \text{ small.}$$

Identify which state(s) can be reached to first order from $|0\rangle$. State the resonance condition on the driving frequency Ω .

Exercise 2 – Consider a neutron of mass m incoming from $z = -\infty$ along the positive z direction.

- (i) Write down the Hamiltonian of the free neutron, the energy eigenvalue and eigenstate for an incoming plane wave with momentum $k_0 \hat{z}$. Give the explicit form of the wavefunction in position space.
- (ii) Near the origin the neutron experiences a spherically symmetric, hard-sphere-type potential

$$V(r) = V_0 > 0 \quad \text{for } r < a.$$

Assume the neutron is very fast, such that its de Broglie wavelength satisfies $\lambda \ll a$. Sketch the expected angular dependence of the scattering amplitude and discuss how many partial waves contribute in this regime.

- (iii) Now assume the neutron is very slow, such that $\lambda \gg a$. Sketch the expected angular dependence of the scattering amplitude, state which partial wave dominates and state whether the scattering is isotropic or anisotropic in this limit.

Exercise 3 – We consider atoms trapped in a one-dimensional infinite square well of length L , defined by $V(x) = 0$ for $0 < x < L$ and $V(x) = \infty$ otherwise. The normalized single-particle eigenfunctions and energies are

$$\varphi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad E_n = \frac{\hbar^2 \pi^2}{2mL^2} n^2, \quad n = 1, 2, \dots$$

We now throw two atoms inside with $\Psi(x_1, x_2)$ denoting the two-particle wavefunction.

- (i) Draw the energy levels and corresponding wavefunctions for a single particle for $n = 1$ and $n = 2$, to scale.
- (ii) Assuming that the atoms are bosons, write the properly symmetrized two-particle ground-state wavefunction explicitly in terms of the single-particle eigenfunctions $\varphi_n(x)$. Find the ground state energy. Introduce bosonic creation operators $\hat{b}_n^\dagger$ and write the $|GS\rangle$ in terms of these operators applied to the vacuum state (no particles).
- (iii) Assuming that the atoms are **spinless** fermions, do the same as above. Introduce fermionic creation operators $\hat{c}_n^\dagger$ and write the $|GS\rangle$ in terms of these operators applied to the vacuum state (no particles).
- (iv) Assuming that the atoms are fermions **with spin** $1/2$, do the same as above. Introduce fermionic creation operators $\hat{c}_{n,\uparrow}^\dagger$ and $\hat{c}_{n,\downarrow}^\dagger$ and write the $|GS\rangle$ in terms of these operators applied to the vacuum state (no particles).
- (v) Assuming that one of the atoms is a boson while the other one is a spin $1/2$ fermion, do the same as above. Use again creation operators to write the ground state.

Possibly useful formulas

pictures of QM Schrödinger:

$$i\hbar \partial_t |\psi_S(t)\rangle = \hat{H} |\psi_S(t)\rangle, \quad \hat{O}_S = \text{const}$$

Time-independent $\hat{H}$:

$$|n(t)\rangle = e^{-iE_n t/\hbar} |n\rangle$$

Heisenberg:

$$\hat{O}_H(t) = e^{i\hat{H}t/\hbar} \hat{O}_S e^{-i\hat{H}t/\hbar}, \quad |\psi_H\rangle = \text{const}$$

angular momentum

$$[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k$$

Expectation values:

$$\langle \hat{O} \rangle = \langle \psi_S(t) | \hat{O}_S | \psi_S(t) \rangle = \langle \psi_H | \hat{O}_H(t) | \psi_H \rangle$$

$$[\hat{L}^2, \hat{L}_z] = 0, \quad \hat{L}^2 Y_{\ell m} = \hbar^2 \ell(\ell+1) Y_{\ell m}$$

Conserved observable:

$$[\hat{O}, \hat{H}] = 0 \Rightarrow \partial_t \langle \hat{O} \rangle = 0$$

hydrogen atom (qualitative)

$$E_n = -\frac{me^4}{2\hbar^2} \frac{1}{n^2}$$

time-dependent Schrödinger equation

$$i\hbar \partial_t |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle$$

Degeneracy (no spin):

$$E_n \text{ independent of } \ell, m$$

Magnetic field:

Zeeman: lifts m -degeneracy

Relativistic corrections:

remove exact n -degeneracy

perturbation theory Time-independent:

$$E_n^{(1)} = \langle n | \hat{V} | n \rangle$$

Degenerate:

Diagonalize $\hat{V}$ in degenerate subspace

Time-dependent:

$$P_{i \rightarrow f} \propto |\langle f | \hat{V}(t) | i \rangle|^2$$

Transitions require:

$$[\hat{V}, \hat{H}_0] \neq 0$$

Resonance:

$$\hbar\Omega = E_f - E_i$$

harmonic oscillator

$$\hat{H} = \hbar\omega(\hat{a}^\dagger \hat{a} + \frac{1}{2})$$

$$\hat{a} = \frac{1}{\sqrt{2}} \left(\frac{\hat{x}}{x_0} + i \frac{x_0}{\hbar} \hat{p} \right), \quad x_0 = \sqrt{\frac{\hbar}{2m\omega}}$$

$$\hat{x} = x_0(\hat{a} + \hat{a}^\dagger)$$

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle, \quad \hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

Ground state:

$$|\psi_0(x)|^2 \propto e^{-x^2/(2x_0^2)}$$

scattering / partial waves Central potential:

channels labeled by ℓ

Low energy:

$$ka \ll 1 \Rightarrow \ell = 0 \text{ dominates (s-wave)}$$

High energy:

$$\ell_{\max} \sim ka$$

Cross sections:

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2, \quad \sigma = \int d\Omega |f|^2$$

Born approximation:

$$|\psi^{(+)}\rangle \approx |\mathbf{k}\rangle + \hat{G}_0^{(+)}(E) \hat{V} |\mathbf{k}\rangle$$

and

$$f(\mathbf{q}) \propto \int d^3r e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r})$$

Asymptotic solution

$$\psi^{(+)}(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k}\cdot\mathbf{r}} + \frac{e^{ikr}}{r} f(\mathbf{k}', \mathbf{k}), \quad \mathbf{k}' = k \hat{\mathbf{r}}.$$

second quantization Bosons:

$$[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}$$

Fermions:

$$\{\hat{c}_i, \hat{c}_j^\dagger\} = \delta_{ij}, \quad (\hat{c}_i^\dagger)^2 = 0$$

Number operator:

$$\hat{n}_i = \hat{a}_i^\dagger \hat{a}_i$$

Non-interacting Hamiltonian:

$$\hat{H} = \sum_i E_i \hat{a}_i^\dagger \hat{a}_i$$

Field operator

$$\hat{\psi}(\mathbf{r}) = \sum_{\alpha} \phi_{\alpha}(\mathbf{r}) \hat{c}_{\alpha}, \quad \hat{\psi}^\dagger(\mathbf{r}) = \sum_{\alpha} \phi_{\alpha}^*(\mathbf{r}) \hat{c}_{\alpha}^\dagger$$

Hamiltonian with interactions

$$\hat{H} = \sum_{\alpha, \beta} h_{\alpha\beta} \hat{c}_{\alpha}^\dagger \hat{c}_{\beta} + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} V_{\alpha\beta\gamma\delta} \hat{c}_{\alpha}^\dagger \hat{c}_{\beta}^\dagger \hat{c}_{\delta} \hat{c}_{\gamma}$$

exchange symmetry Bosons:

$$\Psi(x_1, x_2) = +\Psi(x_2, x_1)$$

Fermions:

$$\Psi(x_1, x_2) = -\Psi(x_2, x_1)$$

Spin- $\frac{1}{2}$ fermions:

$$\text{spatial sym.} \Leftrightarrow \text{spin antisym.}$$